

# Progress Report

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## Research Objective and Goal

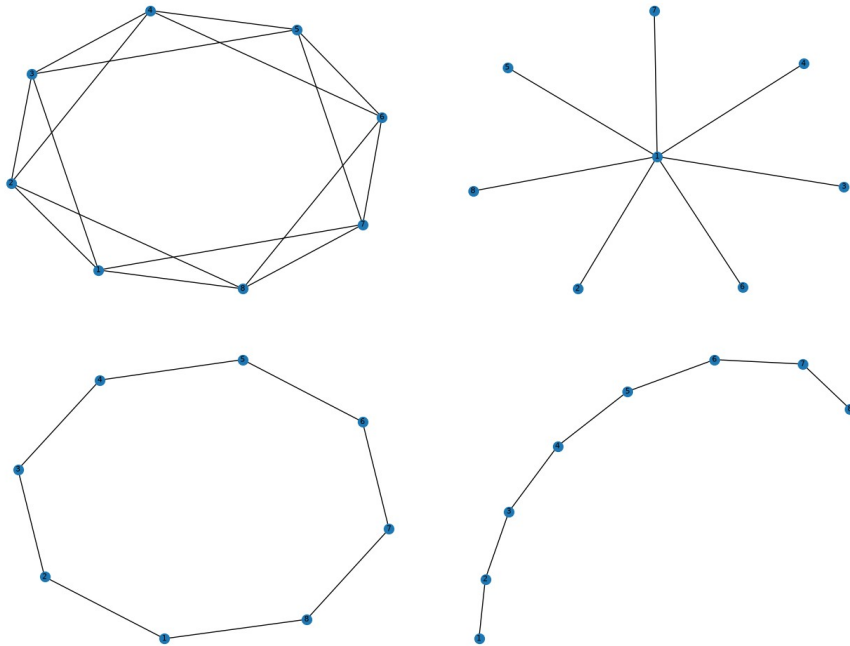
The final goal of the project is to implement the distributed algorithm for PCA (Principal Component Analysis) developed by our lab and test its validity based on numerical experiment.

## Short-Term Goals

10. Continue studying the number of iteration  $T_Y$  and  $T_Z$  in the averaging consensus of the matrices  $Y_m(t)$  and  $Z_m(t)$  and the number of iteration  $T_{PM}$  in the distributed algorithm for PCA alongside with star graphs, path graphs, circle graphs, and regular graphs.

## Progress

We recall that we are currently studying 4 types of graph: the regular graph, the star graph, the circle graph and the path graph. Here's the agent network instances:



We also recall the used global variables:

SEED = 15

CONSENSUS\_EPS = 1e-8

T\_PM = 10000

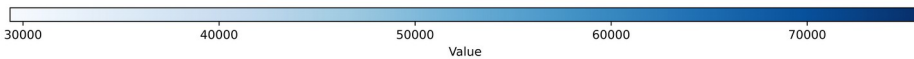
K1 = 0.2, K2 = 0.4, EPS = 0.1

In the last meeting, a table of the number of iterations needed for the decentralized algorithm for PCA to converge to the optimal with a fixed tolerance of error depending on the maximum number of iteration of the inner loop ( $T_Y$  and  $T_Z$ ) were presented.

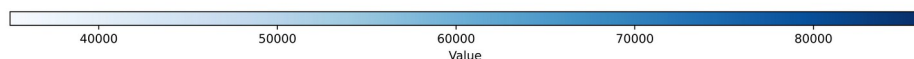
For each graph, here's three new tables while changing CONVERGENCE\_EPS (1e-8, 1e-9, 1e-10).

Regular graph:

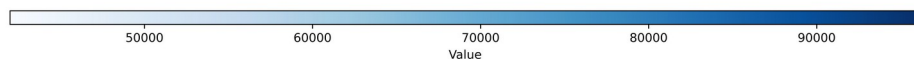
**Number of iterations needed (Total, T PM, avg(T Y), avg(T Z)) | CONVERGENCE EPS=1e-08, regular graph**

[illegible]

**Number of iterations needed (Total, T PM, avg(T Y), avg(T Z)) | CONVERGENCE EPS=1e-09, regular graph**

[illegible]

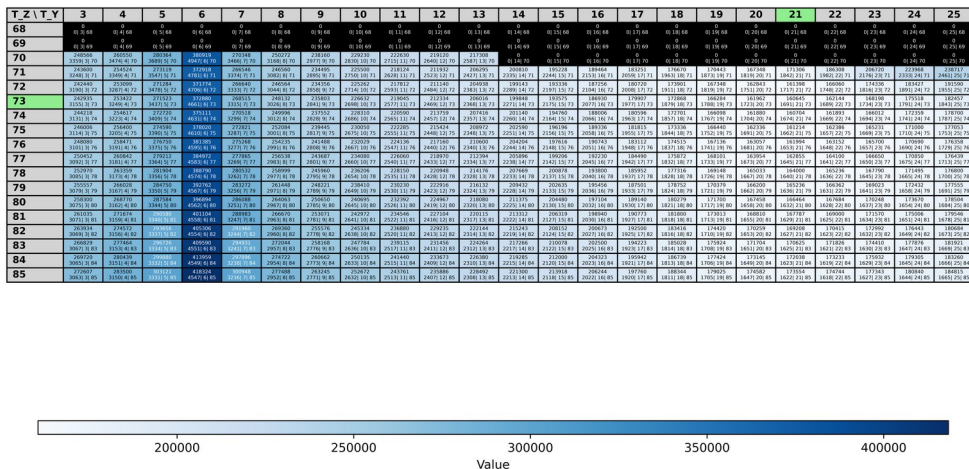
**Number of iterations needed (Total, T PM, avg(T Y), avg(T Z)) | CONVERGENCE EPS=1e-10, regular graph**

[illegible]

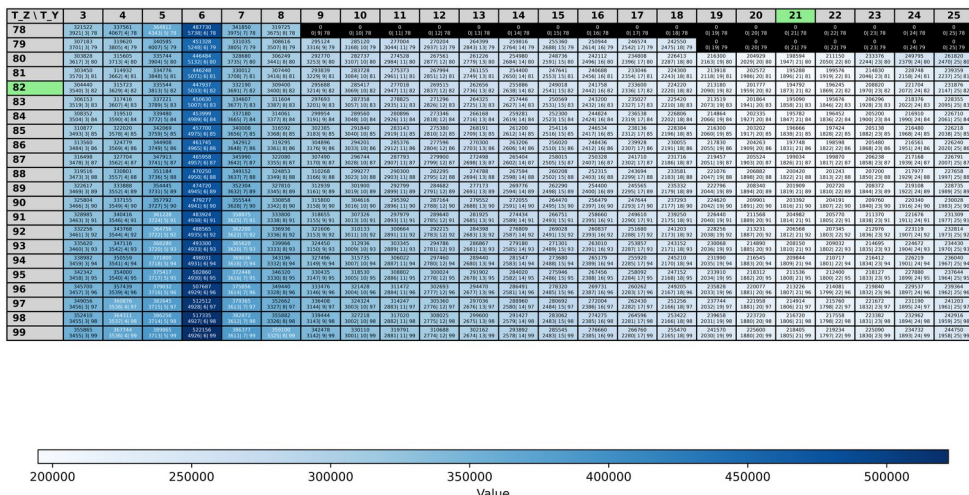


Star graph:

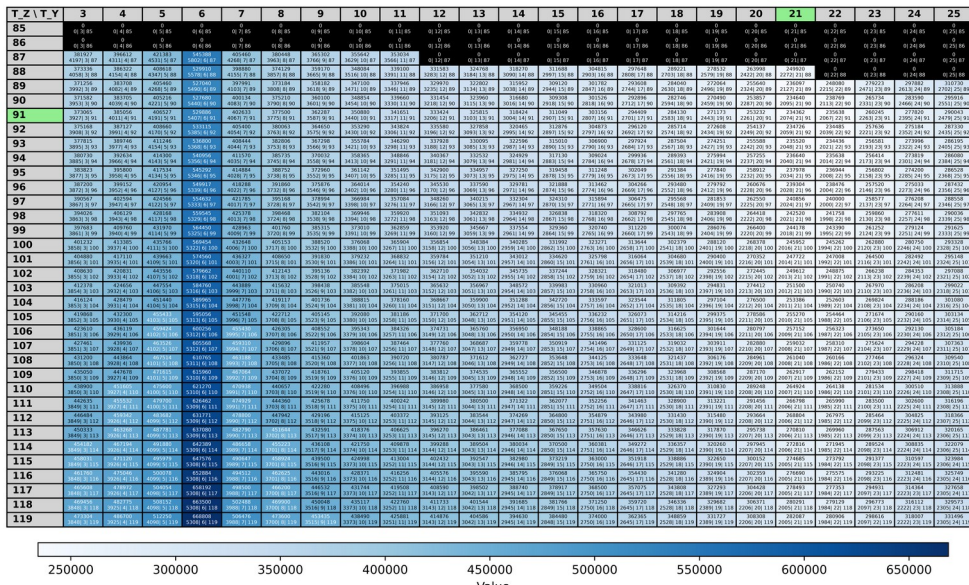
**Number of iterations needed (Total, T\_PM, avg(T\_Y), avg(T\_Z)) | CONVERGENCE\_EPS=1e-08, star\_graph**



Number of iterations needed (Total, T\_PM, avg(T\_Y), avg(T\_Z)) | CONVERGENCE\_EPS=1e-09, star\_graph

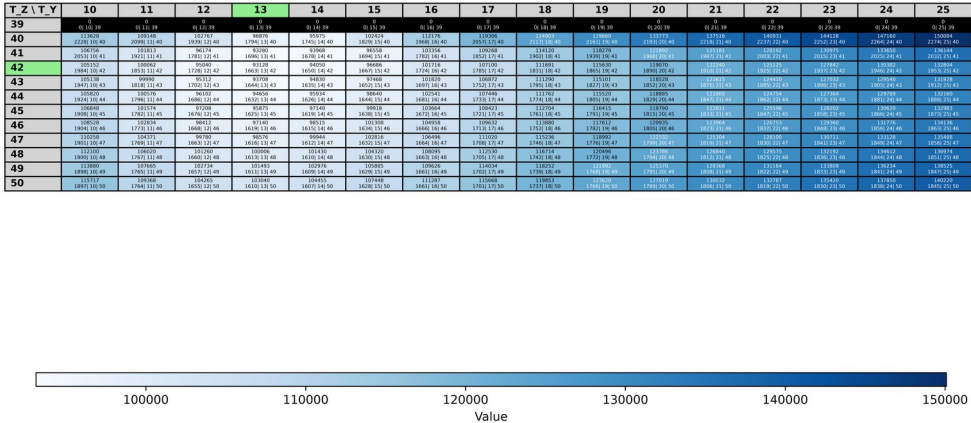


Number of iterations needed (Total, T\_PM, avg(T\_Y), avg(T\_Z)) | CONVERGENCE\_EPS=1e-10, star\_graph

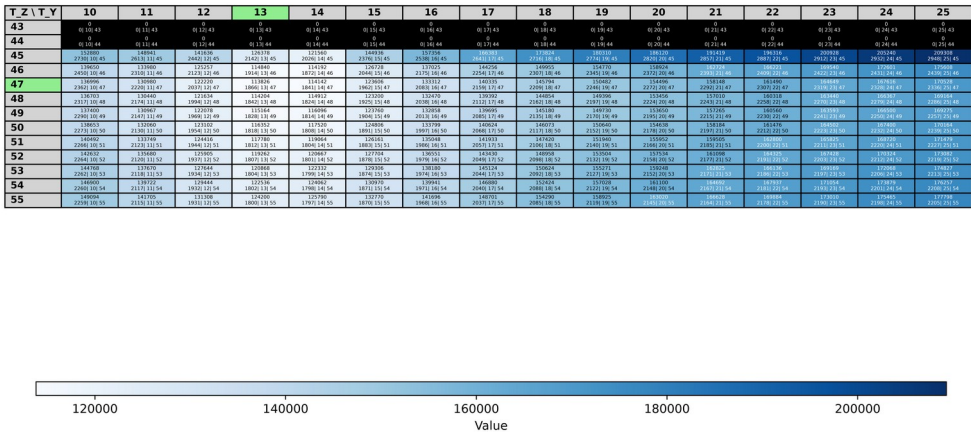


Circle graph:

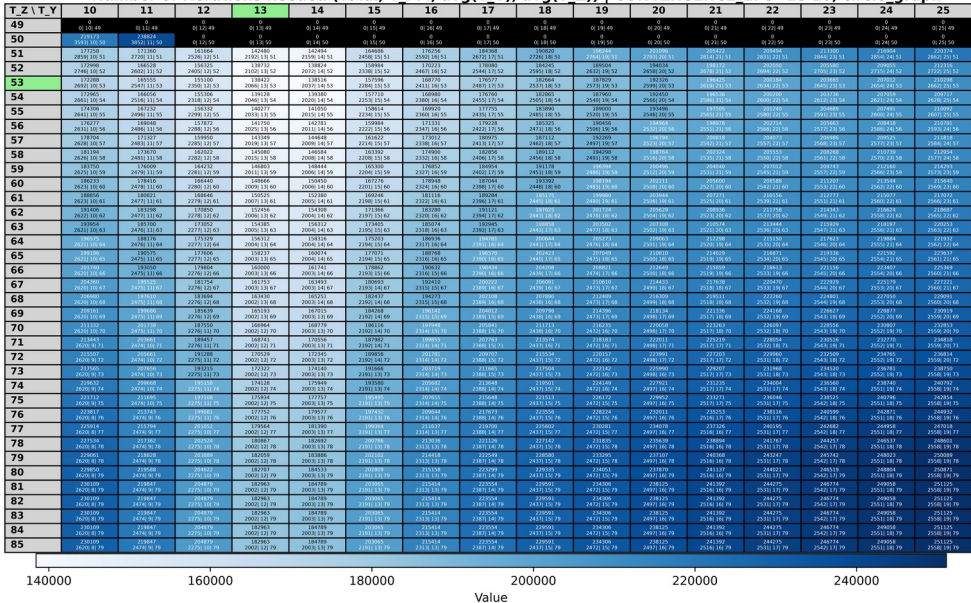
**Number of iterations needed (Total, T\_PM, avg(T\_Y), avg(T\_Z)) | CONVERGENCE\_EPS=1e-08, circle\_graph**



**Number of iterations needed (Total, T\_PM, avg(T\_Y), avg(T\_Z)) | CONVERGENCE\_EPS=1e-09, circle\_graph**



Number of iterations needed (Total, T\_PM, avg(T\_Y), avg(T\_Z)) | CONVERGENCE EPS=1e-10, circle\_graph







As we can see, the stricter the error tolerance, the more iteration to compute matrices  $Z$  we need. As highlighted during previous meetings, we can confirm that the more iterations we perform to compute matrices  $Z$ , the more optimal the convergence limit becomes.

In the progress report 4, we have observed that increasing the number of iterations to compute matrices  $Z$  increase the accuracy of the the convergence limit. But now we observe that increasing the accuracy threshold of the solution increase the number of iterations needed to compute matrices  $Z$ .

In other words, the number of iterations to compute matrices  $Z$  in the averaging consensus algorithm increases if and only if the accuracy of the the convergence limit increases.

(Note that the « Accuracies (at  $1e-i$ ) and average number of iteration needed based on CONVERGENCE\_EPS ( $1e-i$ ) » and the « Accuracies (at  $1e-i$ ) and average number of iteration needed based on CONSENSUS\_EPS ( $1e-i$ ) » tables in progress report 5 are all incorrect due to an improper stopping condition in the algorithms)